

SYSTEMS ARCHITECTURE DEEP DIVE

DeepSeek Harness

The Open-Source Agent Runtime Powered by Cordis

Cyber·X·Lab Technical Research Series | Production Systems Engineering

Production Bottlenecks in Legacy Agent Frameworks

X01

CONTEXT CONTAMINATION

Flat global dictionaries pass unconstrained state. After 20+ steps, dirty variables linger and poison LLM reasoning.

X02

PROCESS TREE LEAKS

Subprocesses treated as raw RPCs. Crashes leave orphaned daemon processes consuming system RAM and socket ports.

X03

RIGID GRAPH COUPLING

Hardcoded static DAG topologies force complete graph rewrites whenever dynamic plugin injection is required.

Goals and Explicit Non-Goals

Architectural Goals

- **Everything as Plugin** via Cordis microkernel slot model
- **Unified Multi-Host Form**: 1 engine powers Web UI (3080), Desktop, CLI & SDK
- **Reversible Effect Streams**: Explicit algebraic effect dual compensation
- **Process Tree Isolation**: Clean POSIX group kill semantics

Explicit Non-Goals

- **No Proprietary Model Lock-in**: Agnostic Model Router supporting DeepSeek & OpenAI specs
- **No Heavy Daemon Bloat**: Avoid forced Docker daemon bundling; delegate to host container contracts
- **No Monolithic RAG Stacks**: Modular plugin boundaries rather than embedded vector pipelines

End-to-End System Topology

HOST LAYER

Unified Clients

- **Web UI:** Port 3080 dashboard
- **Desktop:** Electron app
- **CLI & SDK:** Python bindings

DSH CORE

Cordis Microkernel

- **Event Dispatcher**
- **Context Dependency Engine**
- **Hot-Reload Plugin Mesh**

RUNTIME

Physical Isolation

- **POSIX Process Groups**
- **Virtual Filesystem Jail**
- **Audit & Compensate Log**

Client requests communicate via JSON-RPC / IPC with the microkernel, which coordinates sandbox execution.

Cordis Microkernel: Inversion of Control

Plugin Lifecycle Contract

Every business capability (Model Routing, Tool Sandbox, Session Store) is a standalone plugin:

- `apply(ctx)` : Mount service contracts
- Dynamic dependency declaration via Context
- Zero global mutable variables

Creator Mode: Live Synthesis

When encountering novel tool requirements:

- Agent synthesizes plugin TypeScript code
- In-memory live compilation & validation
- Zero-downtime hot reloading into kernel mesh

Reversible Algebraic Effect Streams

Deterministic State Management

External environment mutations are never executed as uncontrolled side-effects. Every action carries an algebraic dual compensation contract:

1. DISPATCH

FORWARD ACTION

Execute file write, command run, or network mutation with pre-flight snapshot.

2. AUDIT

LEDGER COMMIT

Emit verifiable state delta to append-only effect journal.

3. ROLLBACK

DUAL COMPENSATE

On failure or user undo, execute inverse algebra without leaving dirty files.

POSIX Process Tree Isolation & Orphan Elimination

The Zombie Process Hazard

When an agent executes `npm run dev` or compile jobs, simple `child.kill()` terminates only the immediate shell wrapper:

✗ Child servers remain running in background, occupying ports & memory.

DSH Process Group Solution

- Launch processes with new session ID: `setsid()`
- Subprocesses inherit explicit negative PGID
- Cleanup sends broadcast signal:
`kill(-pgid, SIGTERM)`
- Guaranteed zero orphan leakage across OS resets

Unified Multi-Host Architecture

Web UI

Embedded Vite dashboard serving on `127.0.0.1:3080` for interactive visual debugging, tool inspection & step rewind.

Desktop Client

Electron desktop shell providing native notifications, file drag-and-drop workspace mounting, and local secret vault access.

Headless CLI & SDK

Single binary command-line interface and Python SDK for CI/CD pipelines, SWE-bench evaluations, and automated server tasks.

Identical Runtime Engine: Zero protocol fragmentation across all host environments.

Architectural Taxes & Known Compromises

~3-5%

DISPATCH LATENCY TAX

Dynamic event dispatch through Cordis hooks incurs minor microsecond overhead vs direct monolithic function calls.

I/O

SNAPSHOT JOURNAL TAX

Maintaining reversible algebraic effect logs demands local disk write bandwidth for deep step histories.

POSIX

WINDOWS EMULATION TAX

Process group PGID semantics require specialized Job Object wrappers on native Windows kernels.

Adoption Decision Tree: When to Adopt

✓ Ideal Adoption Scenarios

- Long-running software engineering agents (SWE-bench)
- Dynamic tool synthesis requirements (Creator Mode)
- Multi-environment deployment (Web + Desktop + CLI)
- Mission-critical systems requiring strict rollback safety

✗ Anti-Patterns (Do NOT Adopt)

- Simple single-turn prompt-response QA bots
- Static read-only RAG pipelines without tool side effects
- Ultra-low-latency real-time voice translation relays
- Purely stateless ephemeral serverless lambdas (<500ms)

3-Step Production Quickstart

STEP 01

Installation

Install via package manager:

```
npm i -g @dsh/cli
```

STEP 02

Init Workspace

Bootstrap configuration:

```
dsh init --runtime cordis
```

STEP 03

Launch Host

Start agent runtime:

```
dsh serve --port 3080
```

Verification: Access <http://localhost:3080> to monitor live plugin mesh telemetry and agent execution loops.

OPEN ACCESS & RESEARCH

Cyber·X·Lab

Exploring the Frontier of AI Systems & Sovereign Engineering

 Official Portal
cyberxlab.xyz

 Video Deep Dives
YouTube @CyberXLab

 Real-Time Updates
X / Twitter @CyberXLab